

Revisiting diachronic change in the nominal domain from Latin to modern Romance*

Abstract

The intuition behind this offering is classical and familiar (cf. Schwegerler 1990): morphological richness of nominal inflection underlies the possibility of Left Branch Extraction in Latin (traditionally called hyperbaton) while the loss of morphological richness of nominal inflection yielded the Left Branch Condition that characterizes modern Romance. Specifically, analogous to the notion “strong” and “weak T” in Chomsky 2015, I here adopt the Nominal Strength Parameter (Blümel 2024), a new instantiation of the classical Borer-Chomsky conjecture which localizes syntactic variation in properties of functional heads. It states that the functional nominalizing head n (cf. Borer 2005) comes in two kinds for the purposes of identification of a category label – strong and weak. These two lexical values morphologically correlate with rich (gender, number and case) noun inflection on the one hand, and poor noun inflection on the other. Crucially, the analysis unifies the mentioned syntactic properties pertaining to the distribution of determiner categories with their optionality in Latin and their obligatoriness

*ACKNOWLEDGMENTS TO BE ADDED

in modern Romance (cf. e.g. Longobardi 1994 on Italian). This paper makes the novel proposal that the relevant diachronic change in the nominal domain involved a resetting of the value of n from strong to weak.

1 Introduction

In this paper I reinterpret a classical impression pertaining to the diachrony of Latin towards modern Romance (MR), confining myself here to French, Italian, and Spanish. Generally, this impression ties the loss of morphological richness on various elements to a gradual increase in syntactic rigidity, i.e. a loss of word order freedom in the transition from Latin to MR. Roughly, the transition from what we might descriptively call a non-configurational syntactic profile in Latin gave way to the configurational syntactic systems that characterizes MR. Much recent work has shown that Latin underlyingly is configurational (cf. Devine and Stephens 2006, Ledgeway 2012, Danckaert 2017), i.e. the term non-configurational is taken to be descriptive so as to classify the surface manifestation that distinguishes Latin from MR.

More specifically and focussing on the nominal domain, the intuition draws a connection between the diachronic morphological impoverishment of nouns on the one hand (cf. Pinkster 1990 on the loss of case in relation to the diachrony of prepositions), and two twin syntactic properties of Latin and MR: Latin did not feature articles nor obligatory determiners and allowed for Left Branch Extraction (LBE); with the diachronic loss of case inflection, an article system developed in MR, going hand in hand with

the Left Branch Condition (LBC, cf. Ross 1967). While this is old news, the analysis in which I will couch the described intuition is not. I will implement it by claiming that the nominalizing functional head n (cf. Borer 2005) is parametrized as strong and weak in the sense of Labeling Theory (cf. Chomsky 2013, 2015), building on Blümel (2024) who applies this analysis synchronically. Latin featured the former value while the languages of MR feature the latter value. The distinctive syntactic characteristics, i.e., the distribution of determiner (DET) categories which distinguish Latin and MR follow from this parametrization. Likewise, the lexical parameter morphologically manifests itself in the different degrees of morphological richness of nouns in Latin and MR, specifically in the degree to which gender, number and case forms in the nominal paradigm vary or fail to vary (syncretism). In this sense, this paper supports a traditional view and makes a theoretical contribution, which captures some of the phenomena most prominently captured under the NP-DP-divide (cf. Bošković 2005a *et seq*).

This paper is structured as follows. Section 2 lays out the basic empirical facts to be accounted for. Section 3 presents the theoretical framework and assumptions for the analysis – which is proposed in Section 4. In Section 4 I also remark on some conceptual ramifications. Finally, the paper is concluded in Section 5.

2 Setting the empirical stage

The empirical facts to be captured pertain to the DET-system in Latin and MR. The pattern is simple enough. Needless to say that with this coarse of

be captured by any account: The connection between LBE and morphologically rich concord on the noun and DET to license it – an aspect which some works acknowledge, but few capture as far as the analysis goes.

Latin’s LBE option contrasts with the absence thereof in MR, an array of languages which are subject to the LBC, a term which appeals to the linear position of the category Ross (1967), later updated to include the category immediately dominated by NP by Chomsky (1973, fn. 10), a view which subsequently received a lot of independent and cross-linguistic support, cf. e.g. Grosu (1974). The LBC is anachronistically formulated in (3)² and exemplified by AP-extraction in Italian in (2-a). Of course, the occurrence of AP contiguous with the noun is well-formed, cf. (2-b):

- (2) a. *chi [più alta] fra di loro ha [la t_i autorità].
 who more high among of them has the authority
- b. chi fra di loro ha [la più alta autorità].
 who among of them has the more high authority
 ‘who has the highest authority among them’. (*Italian*,
- Ledgeway 2012, 424

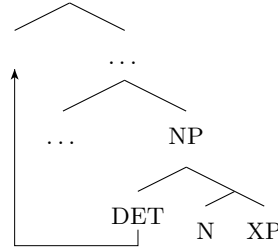
(3) **The Left Branch Condition (LBC)**, Ross (1986, 127)

No NP which is the leftmost constituent of a larger NP can be re-ordered out of this NP by a transformational rule.

To avoid misunderstandings and by way of the tree graph (4), LBE involves extraction of DET in the languages that allow it, and the LBC bans precisely that type of movement in the languages that disallow it:

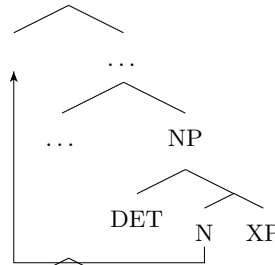
²I retain this formulation here, because it resembles more closely the analysis proposed in this paper, compared to formulations committed to the DP-hypothesis.

(4)

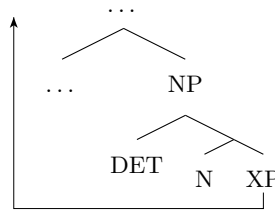


LBE does not designate movement of N as shown in (5-a) or subextraction of XP, the complement of N, as shown in (5-b). At least (5-b) is independently attested in Latin but is plausibly dissociated from the possibility of LBE (e.g. English allows (5-b) but disallows LBE). As far as (5-a) is concerned, whether or not Latin features this type of movement remains to be shown.³ I will briefly get back to extraction of postnominal genitives in Latin in Section 4.2, i.e. an instance of (5-b), but will be primarily be concerned with LBE in this paper.

(5) a.



b.



The island constraint LBC is a deep-seated puzzle, and various attempts to

³I refer the reader to Ott (2011) for compelling arguments that German ‘split topics’ do not involve (5-a), impressions notwithstanding.

derive it exist in the literature. That languages like Latin disobeys the constraint was noticed by Ross (1967) at once. Many researchers have pointed out agreement as one crucial factor involved in the possibility of LBE, i.e., of circumventing the LBC (cf. Ross 1967, Corver 1990, 2010, Bošković 2013a, Reeve 2019). For instance, being concerned with Ancient Greek and using a schematic hyperbaton structure Y_1XY_2 , Devine and Stephens (2000, 224) characterize the relation as follows: “the morphology serves as an overt coindexing system that guides unification of the discontinuous Y elements into a single semantic constituent.” I will elaborate on this aspect in the course of the discussion.

Another well-known observation is that MR encompasses languages which require the presence of a DET-category, i.e. DET is obligatory. While the empirical observation is restricted to singular count nouns, mass nouns and bare plurals are analytically treated along the same lines, i.e., with an obligatory phrase DET which remains silent. The seeming optionality of DET in bare plurals and mass nouns is thus traced to a null-variant of DET in these contexts. In this sense, I follow conventional treatments within the DP-hypothesis, albeit with the qualification that the mass/plural operator is phrasal. The obligatory nature of DET in MR is illustrated by (7)-(9), though it should be noted that French even disallows bare plurals.⁴

⁴The facts must be treated with much care. For example, bare nouns which denote properties of kinds are possible in Spanish as (6) shows.

- (6) a. Lleva jersey.
wears pullover
'(S)he is wearing a pullover.' (It could be one, or more than one)
b. Comprará coche.
buy-FUT car
'(S)he will buy a car.' (It could be one, or more than one) Espinal (2010,

- (7) *(Un/il) Grande amico di Maria mi ha telefonato.
 (a/the) great friend of Maria me have called-up
 ‘(A/the) good friend of Mary called me up.’ (*Italian*, Longobardi 1994, 612)
- (8) Limpio #(el) coche.
 clean (the) car
 ‘I’m cleaning (the) car.’ (*Spanish*, Espinal 2010, 993)
- (9) a. *Etudiants ont occupé l’édifice.
 students have occupied the building
 b. *J’ai vu étudiants dans l’édifice.
 I have seen students in the building
 (*French*, Delfitto and Schroten 1991, 156)

Once more, Latin contrasts with this observation in that DET is optional. This is illustrated by the lack of anaphorically used nouns in (10), where *legatos* ‘emissaries’ introduces the referent and subsequent mentions of that referent must hence be anaphorically definite. No DET-category is required, unlike in the English translation and, importantly, unlike counterparts in MR.

- (10) legatos ad eum mittunt ...legatis respondit. ...ubi ea dies quam
 constituerat cum legatis venit et legati ad eum reverterunt ...
 ‘They sent emissaries to him ... He answered the emissaries ... When
 the day came that he had set up with the emissaries, and the emis-

985)

I abstract from such –and more– complications, like the difference between pre- vs. postverbal positioning of arguments in the Romance pro-drop languages. That there exists an elementary difference between the nominal systems of MR and the one of Latin is not disputed.

saries came back to him ...’ (Caes. B.G. 1.7-8 in Wigtil 1993, 468)

In this connection, it is worth mentioning that Latin lacked articles altogether, a property often associated –but not identical to– with observations like (10). Thus Ledgeway (2012, 424) notes that

“Quintilian’s (Inst. 1.4.19) oft-cited observation ‘our language does not need articles’ might be taken to mean that not only did Latin lack articles, but that it also lacked a dedicated position for D(eterminers).”

When analyzing the observed pattern I will return to an interpretation of the relation between optional DET and the absence of articles in Latin. As before, the link between the availability of articles and the LBC being operative is well-established (cf. cf. Uriagereka 1988, 115; Corver 1990, 2010; Bošković 2005a *et seq*⁵). The resulting one-way correlation at a descriptive level runs as follows:

- (11)
- a. If a language has articles, it is subject to the LBC.
 - b. If a language lacks articles and lacks nominal concord, it is subject to the LBC.
 - c. The only way for a language to be exempt from the LBC is if it lacks articles and exhibits nominal concord.

Now, I take this correlation to be well established. But an important ques-

⁵But see Pankau (2019) for the potential counterexample Lower Sorbian, which however, is a language in close contact with German – possibly a confounding factor.

tion and lacunae is if the absence of articles and the presence of nominal concord are sufficient condition for a language to have LBE. The answer is clearly no. Estonian is a straightforward counterexample since it lacks articles (cf. Norris 2014, 33), exhibits nominal concord with respect to case, but is subject to the LBC, cf. (12)-(13):

- (12) a. Heiko müü-s uue auto maha.
H.NOM sell-PST.3.SG new.ACC car.ACC PRTCL
'Heiko sold a/his new car.'
- b. [Uue auto]_i müü-s Heiko _{t_i} maha.
new.ACC car.ACC sell-PST.3.SG H.NOM PRTCL
'The new car, Heiko sold.'
- c. *Uue_i müü-s Heiko [_{t_i} auto] maha.
new.ACC sell-PST.3.SG H.NOM car.ACC PRTCL
'The new car, Heiko sold.' Norris (2014, 30)
- (13) a. Sa luge-si-d Kärdi raamatu-t.
you read-PST-2.SG Kärt.GEN book-PAR
'You read Kärt's book.'
- b. *Kelle_i sa luge-si-d [_{t_i} raamatu-t]?
who.GEN you read-PST-2.SG book-PAR
Intended: 'Whose book did you read?'
- c. [Kelle raamatu-t]_i sa luge-si-d _{t_i}?
who.GEN book-PAR you read-PST-2.SG
'Whose book did you read?' Norris (2014, 30-31)

Furthermore, the term nominal concord is used in different ways in the literature sometimes encompassing languages which have agreeing modifiers to nouns, where, however, the noun itself does not inflect for the relevant features. German is such a language. (14) illustrates different gender forms on adjectives for the strong adjective declension; (15) illustrates different

case forms for the strong adjective declension:

(14) *Gender (Nominative case)*

- a. ein großer Berg
ein big-M.SG.NOM mountain
- b. eine große Schale
ein big-F.SG.NOM bowl
- c. ein großes Feuer
ein big-N.SG.NOM fire

(15) a. *Accusative*

- (i) einen großen Berg
ein big-M.SG.ACC mountain
- (ii) eine große Schale
ein big-F.SG.ACC bowl
- (iii) ein großes Feuer
ein big-N.SG.ACC fire

b. *Genitive*

- (i) eines großen Bergs
ein big-M.SG.GEN mountain
- (ii) einer großen Schale
ein big-F.SG.GEN bowl
- (iii) eines großen Feuers
ein big-N.SG.GEN fire

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I use nominal concord here in a narrower sense in that next to covariation of certain features on modifiers and inherent ones on the noun, the noun crucially needs to exhibit morphological inflection of the relevant feature. This means that German does not exhibit nominal concord in the intended

sense, because the nouns for the most part fail to morphologically manifest underlying gender and case differences.

I here amend (11-c) by (16) saying that next to the absence of articles, it is the presence of morphologically realized nominal concord on both the noun and DET with respect to the features number, gender and case which is required for the exemption from the LBC:

- (16) The only way for a language to be exempt from the LBC is if
- a. it lacks articles and
 - b. exhibits nominal concord (on N and DET) with respect to number, gender and case.

By naming all the relevant features which need to be involved in nominal concord a more restrictive morphological precondition for LBE is ensured – as is the case empirically and cross-linguistically. Notice that the “if” in (16) lacks an ‘f’ – and deliberately so: It is not a biconditional but weaker. It states that a necessary condition for LBE is the absence of articles and the availability of nominal concord on N and DET which must include all of number, gender and case. Since it is unidirectional, there might still be languages which have all these properties and still be subject to the LBC. Since Estonian is a language which lacks grammatical gender altogether, it does not fall under (16) and so it is expected that the language disallows LBE.

Returning to Latin, a range of categories can undergo LBE in the language. LBE is attested, next to demonstratives as in (1), repeated here as

(17-a), for possessors (17-b), WH-phrases (17-c) (Devine and Stephens 2006, 542ff) and APs (17-d). This is reminiscent of the state of affairs in much of Slavic (cf. Bošković 2005a).

- (17) a. hanc_i cum habeat [_{t_i} praecipuam laudem].
 this.ACC since has particular.ACC merit.ACC
 ‘since he has this particular merit’ (Cicero, Brutus 261)
- b. meo_i tu epistulam dedisti [_{t_i} servo].
 my.DAT you.NOM letter.ACC gave servant.DAT
 ‘you gave the letter to my servant?’ (Plautus, Pseudolus 1203)
- c. quis_i umquam [_{t_i} Graecus] comoediam scripsit
 which.NOM ever Greek.NOM comedy.ACC write
 ‘Which Greek ever wrote a comedy?’ (Cicero, pro Flacco 27.65)
- d. qui **summam**_i inter eos habet [_{t_i}
 who highest.ACC.F.SG among them.ACC has
auctoritatem].
 power.ACC.F.SG
 ‘who has the highest authority among them’ (Caes. B.G. 6.13.8; in Bauer 2009, 290)

The paradigm in (18)-(20) gives illustrations of the LBC in MR.

(18) *French*

- a. *Quels_i a-t-il vendu [_{t_i} livres]?
 which has-he sold books
 ‘Which books did he sell?’ (in Corver 2010, 6)
- b. *Cette_i j’aime [_{t_i} fille]!
 that I love girl
 ‘That girl I love!’ (in Corver 1990, 19)

- c. *Son_i j'ai lu [_{t_i} livre]!
his I have read book
'His book I have read!' (in Corver 1990, 19)
- (19) *Questi_i hai incontrato [_{t_i} ragazzi]?
which you-have met boys
'Which boys did you meet?' (*Italian*, Rizzi 1982)
- (20) *¿Cuántos_i compró María [_{t_i} libros]?
how many bought Mary books
'How many did Mary buy books?' (*Spanish*, Francom 2012, 540)

So an important task is accounting for the change that led from Latin to MR in the nominal domain.

Bošković (2005a) takes the possibility of AP-extraction to be definitional of LBE, and I follow him in this regard. There seems to be an implicational relation between the possibility of AP-LBE and the other DET-categories. That is, I have not found a language which allows AP-LBE but not also all the other DET-categories (wh-determiners, demonstratives, possessors, etc.). The reverse is not true in that languages which allow selected DET-items to undergo extraction never at the same time allow AP-LBE. Thus, Hungarian famously permits extraction of dative possessors (Szabolcsi 1983) but disallows APs to do the same. Much of Germanic allow the extraction of the counterpart *what* in *what for a book* ('what kind of book', cf. Leu 2008, 2014 and references therein) but disallows APs to do the same. French *combien* ('how many') can split off from the associated noun phrase, but APs cannot (cf. Starke 2001). And so on.

While I subscribe to the view that AP-extraction define LBE-languages (while languages of the exceptional type I just mentioned are not defined

this way), a caveat is required: I here abstract away from the analytical treatment of APs, but will here be concerned with the analysis of the other DET-categories illustrated. The reason is that even though AP-extraction makes the right demarcation at a descriptive level, their analysis poses two non-trivial problems with rich conceptual and empirical ramifications: Their mode of being introduced into structure (pair-Merge or set-Merge, cf. Oseki 2014 for an important take at unifying the two) on the one hand and nominal concord for attributive adjectives on the other (for one approach, cf. Norris 2014). The analytical treatment of APs exceeds the limits of this paper, but I hope to address it in the future.

Summarizing, we have seen that DET occurs optionally in complex nominals in Latin and obligatorily so in MR. Moreover, Latin allows LBE of DET-categories but MR does not. Descriptively, we can conclude that external and internal Merge of DET is optional in Latin, while external Merge is obligatory and internal Merge of DET is banned in MR. Before presenting the analysis of this pattern, I will lay out the theoretical background in which it is developed.

3 Theoretical background: Labeling Theory

Recent syntactic theorizing is characterized by an effort to reduce the ingredients of the operation which builds hierarchical structure called Merge (cf. Chomsky 1995). These efforts have arguably reached their peak, reflected in the superlative of the term Simplest Merge, cf. Epstein et al. (2014). A parallel development has moved towards progressively liberating the ap-

plication of Merge from the dominant role that features play in syntactic derivation (Chomsky 2004, 2013, 2015, the latter two works will be referred to as POP/+). A fundamental assumption that has emerged from these tendencies of inquiry is that the application of Merge is not feature-driven – while the representations emerging from its application are to an extent subject to conditions which lexical properties determine. Specifically, (both external and internal) Merge applies freely, untriggered, as shown schematically in (21), which is also meant to illustrate (a) that no linear ordering is imposed on the elements α and β and (b) that no projection mechanism is built into the operation.

$$(21) \quad \text{Merge}(\alpha, \beta) \rightarrow \{\alpha, \beta\}$$

Of course, for this view to be tenable descriptively, there must be constraining factors to prevent overgeneration. By assumption, these factors are extraneous to narrow syntactic computation. Possibly and as suggested in POP/+, these factors apply only at the point when the cyclic operation Transfer hands over syntactic representations to the interfaces in a phase-by-phase fashion. The specific operationalizations that POP/+ proposes are given in (22-a) and (22-b):

- (22) a. the requirement that syntactic objects be labeled, informing the question: What kind of object is it?
- b. the 3rd factor principle of computational efficiency *Minimal Search*, which detects labels in an efficient manner, i.e., structurally closer is better than remote (POP: 43).

I will henceforth refer to (22-a) as the Labeling Requirement LR. To reiterate: Merge is maximally simple, i.e., just (21). It symmetrically combines two elements within an unordered set and without any encoding of projection and thus no notion of specifier.

To meet the LR, POP proposes a Labeling Algorithm (LA) which finds the most prominent element in a given set. The guiding intuition behind the LA is that structurally closer wins out over structurally remote. Whenever a simple element Merges with a complex one, the simple one wins out. When the two Mergees are equally complex, more needs to be said. The three logically conceivable cases below illustrate the LA with specific examples. Underlining is used to indicate the element found as a label:

Head-XP: LA detects the head as the label (XP requires deeper search).

$$(23) \quad \{V, NP\} = \{V, \{N, \dots\}\} \rightarrow \{\underline{V}, NP\} = VP$$

Head-Head: The label is the categorizer which bears grammatical features, not the root R, which does not bear grammatical features and is thus “too weak” to be found as a label.

$$(24) \quad \{v, R\} \rightarrow \{\underline{v}, R\} = vP$$

XP-YP: The label is either

- YP iff XP undergoes internal Merge (i.e., moves)
E.g., intermediate configurations in successive-cyclic movement of the format

$$(25) \quad \{\alpha \text{ XP, CP}\} \rightarrow \text{XP} \dots \{\text{CP} \langle \text{XP} \rangle, \{\underline{\text{C}} \dots\}\}$$

- or $\langle f, f \rangle$, where X and Y bear f and agree with respect to f .

(26) the target of EPP-raising in English:

$$\{\text{NP, TP}\} \rightarrow \{\langle \varphi, \varphi \rangle \{\text{N}_{\underline{\varphi}} \dots\}, \{\text{T}_{\underline{\varphi}} \dots\}\}$$

The option (26) implicates an innovation compared to classical $\bar{\text{X}}$ -theoretic and Bare Phrase Structure-conceptions of projecting a category tag: The LA can target features, not only Lexical Items.⁶

The derivation of a simple finite sentence comprising a transitive verb phrase in (27) illustrates the working of the LA, including EPP-raising of the subject-NP. For example, *Susan kicks the ball*, where NP=*Susan*, $v\text{P}$ =*kick the ball* and T =[Tense: PRS][$\text{u}\varphi$:_]:

- (27) a. $\{\text{?NP, } v\text{P}\}$ labeling problem; (external) Merge T
b. $\{\text{T, } \{\text{?NP, } v\text{P}\}\}$ (internal) Merge NP
c. $\{\text{NP, } \{\text{T, } \{v\text{P} \langle \text{NP} \rangle, \{v, \dots\}\}\}\}$ labeling possible
d. $\{\langle \varphi, \varphi \rangle \text{ NP, } \{\text{T, } \{v\text{P} \langle \text{NP} \rangle, \{v, \dots\}\}\}\}$

In (27-a) there is a problematic symmetry in that neither N nor v can provide the label of NP- $v\text{P}$. Assuming that NP raises upon Merge of T, NP- $v\text{P}$ receives a label by v , because by assumption, the lower copy of the subject-NP is invisible to LA. NP-TP in (27-c) is labeled by the common feature

⁶Cf. POP: 45: “it must be that LA seeks features, not only LIs – or perhaps seeks only features, in which case it would be similar to probe-goal relations generally, specifically Agree.” The current approach to NCs extends this line of thinking.

borne by N and T, namely φ . The result is shown in (27-d). This gives the desired results whenever a transitive structure is involved.

What of a *general* deduction of the EPP-property in English, including the obligatory raising of grammatical subjects in, e.g., passives and unaccusatives? To solve this problem, POP+ introduces the notion of “weak T” and “strong T.” The intuition behind this distinction is that there the morphological richness of verb inflection signals an underlying lexical parameter of T with significant consequences for the syntactic computation and the LA. Thus weak T underlies the poorly inflecting verbal φ -features in English. Weak T is in effect a defective category to label α in (28-a). Raising of the internal argument needs to apply to “support” weak T so that a $\langle\varphi, \varphi\rangle$ -label can be identified by the LA as in (28-b):

- (28) a. $*[\alpha T_{wk}=was [kicked\ the\ ball]]$
 b. $[\langle\varphi, \varphi\rangle [NP\ the\ ball]_i T_{wk}=was [kicked\ t_i]]$

As before the feature shared between N and T results. This time, however, EPP-raising obtains not for a problematic symmetry in the base structure but because of the defective nature of weak T to provide the label for its containing set.

POP+ shows that this conception makes a unification of the EPP and the Empty Category Principle (ECP) possible. Consider the *that*-trace effect in (29). From the perspective of Labeling Theory, $\bar{A}$ -moving a subordinate subject-NP restores the problematic unlabelability of α in (29) due to the invisibility of the “trace” for the LA:

- (29) **Who_i does Benjamin think that [_{α} t_i T_{wk} kicks the ball]*

Unlike English, Italian with its rich verbal agreement paradigm has “strong T.” As a consequence, there is no need (a) to raise an NP to the sister position of TP, and (b) to retain the subject within the subordinate clause. The uniform reason is that strong T can label the structure without support. This, according to POP+, is the underlying cause for pro-drop (30-a) and for the absence of ECP-effects (30-b) in Italian and comparable pro-drop languages:

- (30) a. [_{*TP*} (*Voi*) T_{*str=state*} [_{*vP*} leggendo un libro]]
 (you) are reading a book
(D'Alessandro 2015)
 b. *Chi_i credi che* [_{*TP t_i*} T_{*str*} *vincerà*] win-will
 who think_{2.SG} that
 ‘Who do you think will win?’ (Rizzi and Shlonsky 2007)

The upshot of this presentation is this: Based on the freely applying operation Merge and in interaction with 3rd factor principles and interface requirements, diverging syntactic effects derive from a single parametric property on the functional head T: its strength and weakness.

This should suffice as theoretical background and we are now in a position to develop the current analysis. In a nutshell, I propose that the conception of parametrizing T for the purposes of labeling can be fruitfully carried over to the nominal domain.

4 Proposal: nominal strength and weakness

In this section I suggest that analogous to the strong/weak distinction on T, the nominalizing functional head n should be parametrized. That is, we expect to find nominal strength and weakness. Let us briefly go back to the previously mentioned common observation that LBE goes hand in hand with rich morphology while the LBC does not: “As was already hinted at by Ross [1967], languages violating the LBC are characterized by rich morphology” (Corver 2010). The observation raises the question which element it is which exhibits “rich morphology.” The central hypothesis in Blümel 2024 to appear is given in (31), and I apply it here to the languages listed:

(31) **The Nominal Strength Parameter (NSP)**

- a. strong n/n_{str} : Latin
- b. weak n/n_{wk} : French, Italian, Spanish . . .

Implied is the conception of functional heads which categorize category-neutral roots (cf. Marantz 1997, Borer 2005), here n .

What is the empirical motivation for parametrizing n ? The classical intuition going back to at least Schwegler (1990) is that the morphological richness of Latin nominal inflection is tied to its LBE characteristic. More specifically, the gradual change in nominal morphology concerned gender and case inflection. Classical Latin (the Old Latin period until around 75 BC) had the three values masculine, feminine and neuter within its gender system and lost neuter in later stages, often associated with the term Vulgar Latin, having both historical and sociological aspects. The remaining femi-

nine/masculine gender system on nouns permeated MR.⁷ Likewise, the rich case system of Latin nouns exhibited –depending on the exact counting– 5-8 values with at least nominative, accusative, dative, genitive and ablative, and possibly vocative, locative and instrumental in addition. These were lost on nouns entirely in MR.

The loss of morphological richness of nominal inflection yielded the LBC that characterizes MR. Abstracting away from declension classes, let us rehash the nominal inflection and case paradigms in Latin, according to (31-a) a strong *n* language, and compare it the morphological properties of the MR languages which are set to the weak side of the parameter, cf. (31-b). The comparison is straightforward as the table in (32) shows in relation to (33):

(32) **The strong**

Latin, sg.	1.F	2.M	2.N
NOM	<i>-a</i>	<i>-us</i>	<i>-um</i>
GEN	<i>-ae</i>	<i>-i</i>	<i>-i</i>
DAT	<i>-ae</i>	<i>-o</i>	<i>-o</i>
ACC	<i>-am</i>	<i>-um</i>	<i>-um</i>
ABL	<i>-a</i>	<i>-o</i>	<i>-o</i>

(33) **The weak**

- French: No nominal case inflection

⁷But see Loporcaro (2017) for a more nuanced and comprehensive view.

- Italian: No nominal case inflection
- Spanish: No nominal case inflection

Against the backdrop of the observations in the sentential domain in Section 3, we can establish the following descriptive claim, assuming that the obligatory presence of DET-categories is an EPP-effect:

- (34)
- a. The richness of nominal inflection is the nominal counterpart of the richness of verbal inflection. Just as the latter licenses (a) pro-drop and (b) circumvents ECP-effects, the former licenses (a') DET-drop and (b') circumvents ECP-effects.
 - b. Poverty of noun inflection is the nominal counterpart of the poverty of verbal inflection. Just as the latter gives rise to a (a) EPP-effect and (b) ECP-effects in the clause, the former gives rise to (a') EPP-(=obligatory DET-)effects and (b') ECP-effects in the nominal domain.

4.1 Executing the analysis and initial predictions

In line with the head-head condition in (24), a minimal NC is (35):

- (35) $\{n, R\}$

R is a category-neutral root, constituting the open class elements. Depending on the functional head R Merges with, it will be categorized differently:

- (36) a. the big $[n \sqrt{house}]$

- b. They [$v \sqrt{house}$ me].

n is a nominalizing functional head (cf. Marantz 1997, Borer 2005). I will assume that it bears gender, number and Case features.

$$(37) \quad n_{[GEN, NUM, CASE]}$$

An additional assumption of the analysis is that determiners are phrasal (Sag et al. 2003, Chomsky 2007, 25-26, Oishi 2015). For the purposes of this paper and following Leu (2008, 2014), I will assume that DET-categories are adjectival (xAP, extended adjectival phrase), headed by xA.⁸ As an effect, Merging [n R] and xAP yields (38), an instance of XP-YP:

$$(38) \quad \{\text{xAP}, \{n, \text{R}\}\}$$

Without any additional assumptions, the NSP makes the following six predictions pertaining to the distribution of DET categories in Latin and MR:

$$(39) \quad n_{str} \text{ (Latin)}$$

- a. [$_{\alpha} n_{str}$ R]
- b. [$_{\alpha}$ xAP [n_{str} R]]
- c. xAP _{i} . . . [$_{\alpha} t_i$ [n_{str} R]]

$$(40) \quad n_{wk} \text{ (MR)}$$

- a. $^*[_{\alpha} n_{wk}$ R]
- b. [$_{\alpha}$ xAP [n_{wk} R]]

⁸Generalizing a view that was initially confined to Slavic, cf. Horn (1978, 1983), Zlatić (1997), Bošković (2005a).

c. *xAP_{*i*}... [_{α} *t_i* [*n_{wk}* R]]

The LA labels α in (39-a) by n_{str} which empirically translates into the observation that in Latin, determiners are optional. The LA labels α in (39-b) by a feature n_{str} and xA share. This will accommodate an aspect of nominal concord in Latin as I will elaborate momentarily. The LA labels α in (39-c) by n_{str} which captures the familiar observation that LBE is possible in Latin. As can be seen, all structures meet the LR, because n_{str} supports a label throughout. After having considered the predictions as far as the n_{str} language Latin goes, let us consider the n_{wk} languages of MR.

The LA cannot label α in (40-a) because n_{wk} and R are too weak to be detected by the LA. This captures the observation that MR obligatorily requires the presence of a DET category, now interpreted as a violation of the LR. The LA labels α in (40-b) because n_{wk} receives “labeling support” by xAP: They share a prominent feature. This is analogous to what POP+ suggests for the EPP in English-type languages. Finally, the LA has to leave α in (40-c) unlabeled because n_{wk} and R are too weak to label and the trace is invisible to LA. This aspect of n_{wk} derives the LBC in MR from the LR. The next subsection addresses the question which respective labels obtain.

As we have seen, core data from the distribution of DET that distinguished Latin from MR are captured by parametrizing n : n_{str} in Latin yields optional DET and LBE, while n_{wk} in MR yields obligatory DET and the LBC. The diachronic passage from Latin to MR which suggests itself is thus one in which the parameter is turned from strong to weak with the concomitant syntactic effects:

$$(41) \quad n_{str} > n_{wk}$$

This resetting of the parameter of n went hand in hand with a loss of gender and case inflection on nouns – a well-established observation in the philological literature. Of course, the idea is that this loss of case inflection was a morphological reflex of the lexical change embodied in (41).

Now let us refine both predictions and analysis.

4.2 Further predictions and refinements of the analysis

Ledgeway (2012) mentions other important ways in which Latin differed from MR in the nominal domain. He observes that placement of possessive adjectives can be prenominal (42-a) and postnominal (42-b) in Latin.

$$(42) \quad \begin{array}{ll} \text{a.} & \text{Caesar} \quad [\alpha \text{ suas} \quad \text{copias}] \quad \text{subducit} \\ & \text{Caesar}_{NOM} \quad \text{his}_{ACC.FPL} \text{ troops}_{ACC.FPL} \text{ withdraws} \\ & \hspace{15em} (\text{Lat., Caes. B.G. 1.22.3}) \end{array}$$

$$\begin{array}{ll} \text{b.} & [\alpha \text{ copias} \quad \text{suas}] \quad \text{Caesar} \quad [\dots] \text{ subducit} \\ & \text{troops}_{ACC.FPL} \text{ his}_{ACC.FPL} \text{ Caesar}_{NOM} \quad \text{withdraws.} \\ & \hspace{15em} (\text{Lat., Caes. B.G. 1.24.1}) \end{array}$$

$$(43) \quad \begin{array}{ll} \text{a.} & \text{César retire} \quad [\alpha \text{ ses troupes}]. \\ & \text{Caesar withdraws} \quad \text{his troops} \\ & \text{'Caesar withdraws his troops'}. \\ \\ \text{b.} & * \text{César retire} \quad [\alpha \text{ troupes ses}]. \\ & \text{Caesar withdraws} \quad \text{troops his} \\ & \text{'Caesar withdraws his troops'}. \hspace{10em} (\text{French}) \end{array}$$

The current approach captures this in that n_{str} can invariably label α – it thus does not matter if the possessor appears in “SPEC” of n as in (42-a)

or what is arguably a complement position of the possessee noun in (42-b). This contrasts with MR, where possessives must show up in “SPEC”- n_{wk} to support n_{wk} for the identification of α ’s label, as exemplified by (43-a) and (43-b). That is, the deviance of (43-b) derives from the structure’s violation of the LR.

Just what *is* the label of α in (39-b) and (40-b)? Analogous to the approach in Chomsky (2015) for the TP-domain, pairings of features f borne by xA and n suggest themselves as shown in (44-b) and (45-b):

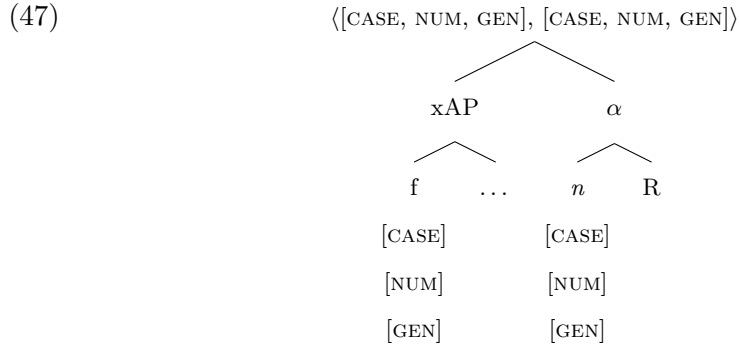
- (44) a. $[\alpha=nP \ n_{str} \ R]$
b. $[\alpha=\langle f,f \rangle \ xAP \ [n_{str} \ R]]$
c. $xAP_i \dots [\alpha=nP \ t_i \ [n_{str} \ R]]$
- (45) a. $*[\alpha=? \ n_{wk} \ R]$
b. $[\alpha=\langle f,f \rangle \ xAP \ [n_{wk} \ R]]$
c. $*xAP_i \dots [\alpha=? \ t_i \ [n_{wk} \ R]]$

Latin exhibits nominal concord in case, gender and number as shown by the AP *instabilis* ‘unstable’ and associated noun *fors* ‘fortune’ in (46-a) and by the demonstrative *hic* ‘this, here’ and the associated noun *splendor* ‘brightness’ in (46-b).

- (46) a. **In-stabilis** in istum plur-imum
NEG-stable.F.SG.NOM in this.M.SG.ACC much-SUPL
fortuna valuit.
fortune.F.SG.NOM strong.3.SG.PERF.IND.ACT
‘Unstable fortune has been most powerful against this one.’
(Rhet. Her. 4.44 in Feldcamp 2021)

- b. **Hunc** tu vitae **splendorem**
 this.M.SG.ACC 2.SG.NOM life.F.SG.GEN brightness.M.SG.ACC
 maculis aspergis istis?
 stain.F.PL.ABL sully.2.SG.PRES.IND.ACT this.F.PL.ABL
 ‘You sully this brilliance of a life with those insults?’

This suggests that the relevant label f of the noun group could be either or all of these features, most of which are abstract in n_{wk} -languages. I will assume so such that the relevant labels are as in (47):



Importantly, these features are not stipulated diacritics but reflect the empirical facts (if the evidence for abstract Case is indirect in some of the languages under consideration). What is a label needed for? I presume that labels are required for interpretation at the interfaces: For example, unordered sets generated by Merge need to be linearized when externalized, and for determining the directionality of the head a label could be required. As for semantic interpretation (47) raises the question what role Case and gender play, since they are both uninterpretable. I will assume that the identification of the label is a requirement for semantic interpretation, but not that the label itself needs to be interpreted. As far as Case goes, this is

reminiscent of the Visibility Condition which states that an element needs to bear Case in order to be visible for θ -marking (Chomsky 1986). (47) could be considered a reinterpretation of this condition: A nominal set formed by Merge needs a label to be visible for θ -marking. Case (and gender and number) is such a label.

To once more highlight another important aspect, it has been commonplace that nominal concord or agreement of sorts is a precondition for LBE essentially since Ross (1967). However, morphologically manifested nominal concord obtains even in MR in terms of number and gender, clearly showing that in and of itself, nominal concord is not enough for the licensing of LBE:

- (48) a. l-a plante vert-e
 the-F.SG plant.F.SG green-F.SG
 ‘the green plant’
- b. l-e bâtiment vert
 the-M.SG building.M.SG green.M.SG
 ‘the green building’ (*French*, Norris 2017)

What seems to be required is rich noun inflection – the crucial property that distinguished Latin from MR. The current approach in terms of n_{str} and n_{wk} accommodates these facts naturally in that the former underlies a rich and varied nominal case paradigm while the latter underlies an absent case paradigm.

In his discussion on discontinuous noun phrases, Pinkster (2021, 1098) recently remarks that discontinuity is not contingent on agreement as evidenced by examples like (49) where a postnominal genitive attribute undergoes fronting, which does not agree with the head noun:

- (49) Quis enim est qui putet . . . **nostrorum hominum** ad eorum doctissimas voces **aures** clausas fuisse?
- ‘For who can imagine that the ears of our countrymen were closed to the echo of their wisdom?’ (Cic. Tusc. 4.3.)

A remark is in order, and a question. The example (49) is not counterevidence to the claim made in this paper that LBE is licensed by rich agreement generally, and rich noun inflection in particular. As Pinkster (2021, 1098) points out, the NP *nostrorum hominum* shows up postnominally in the contiguous variant and thus does not fall under the rubric LBE. Secondly, why Latin allowed postnominal genitive attributes to be extracted from NPs is an interesting question. Languages appear to vary with respect to this possibility. Thus Serbo-Croatian (50) and German (51) disallow it, while Czech marginally allows it (52):

- (50) a. ?*[Ovog studenta]_i sam pronašla [knjigu _{t_i}]
 this.GEN student.GEN am found book.ACC
 ‘Of this student I found the book.’
- b. *Koga_i si pronašla [knjigu _{t_i}]
 who.GEN are found book.ACC
 ‘Of whom did you find the book?’ (Zlatić 1997 in Bošković
 2013b)
- (51) a. *[Dieses Studenten]_i habe ich [ein Buch _{t_i}] gefunden.
 this.GEN student.GEN have I a book found
- b. *Wessen_i habe ich [ein Buch _{t_i}] gefunden.
 whose.GEN have I a book found
- COLLEAGUE, PERSONAL COMMUNICATION

- (52) $?(?)$ [Kojeg doktora]_i si vidio [prijatelja t_i]
 which doctor.GEN are seen friend.ACC
 ‘Which doctor did you see a friend of?’ (Starke 2001, 39 in
 Bošković 2018)

The emerging picture with respect to the extraction of complements of nouns is inconsistent: The article-less languages Serbo-Croatian, Czech and Latin exhibit rich noun inflection but differ with respect to extracting postnominal genitive NPs. The article-language German has poor nominal inflection and appears to pattern with Serbo-Croatian.

Example (52) involves extraction of the genitive NP from a structurally case marked NP and is chosen to contrast with an inherently case marked NP. The judgment is thus possibly to be taken within this context and arguably needs to be idealized. Concluding that the richness of nominal inflection does not figure in the licensing of this possibility is thus premature. Richer and more careful data elicitation and argumentation is required. Why Latin appeared to permit extraction of genitive NPs remains to be understood.

4.3 A discussion

Let us take stock. In this paper, I suggest a uniform analysis of the distribution of DET-categories in Latin and MR couched in the POP/+ framework (Chomsky 2013, 2015). The analysis has the following characteristics: Merge applies freely and untriggered, and there is no projection. Instead, an interface condition on labeling (the LR) and 3rd factor principle of efficient computation interact to deliver the label for a given set formed by Merge.

Following Blümel (2024), I have adopted the Nominal Strength Parameter which states that n is parametrized as strong (Latin) and weak (MR). The diachrony is related to the loss of morphological nominal inflection, indicating the underlying replacement of strong n by weak n . The approach has many implications of which I can address only one here, diverting a fuller discussion to a different occasion (AUTHOR in progress); I will confine myself to diachronic aspects in this discussion.

At least as far as the diachrony of the syntax of the nominal structure in Romance goes, the current approach casts doubt on economy principles that attribute historical clines on preferences of head over phrase status as forcefully suggested in much recent work by van Gelderen (2004, 2018, 2019) and Dadan (2019). van Gelderen (2004, 18) suggests the Head Preference Principle given in (53):

(53) **Head Preference Principle**

Be a head, rather than a phrase.

A commonality of these works is that *Minimal Search* exerts a pressure on structures of the form XP-YP such that over time, they gave way to Head-XP structures. In the nominal domain, this is played out by demonstratives turning into articles over time. By assumption, the former are phrasal and the latter are heads. Thus, there was a change from demonstratives in Old English (a language lacking articles) towards stages in which it assumed an article system over time, illustrated in (54).

(54) a. Brittene igland is ehta hund mila lang. & twa hund brad. &

her sind on **þis** iglande fif geeode. Englisc. & Brittisc. & Wilsce.
& Scyttisc. & Pyhtisc. & Boc Leden.

‘The island Britain is 800 miles long and 200 miles broad. And
there are in the island five nations; English, Welsh, Scottish,
Pictish, and Latin.’ Old English-style in the Peterborough
Chronicle

- b. is gære for **þe** king Stephne ofer sæ to Normandi; and ther wes
underfangen forþi at hi uuenden at he sculde ben alsuic alse
the eom wes,
‘This year, (the) King Stephen crossed the sea to go to Nor-
mandy and was received there because they thought he was like
the uncle.’ 1137

van Gelderen (2019) adumbrates the diachronic development shown in (55)
for this change; the DP on the left side of the arrow represents the demon-
strative.

$$(55) \quad [? [_{DP} D] [_{NP} N \dots]] \rightarrow [_{DP} D [_{NP} N \dots]]$$

The former is an XP-YP structure while the latter is a Head-XP structure.
The former allegedly sits uneasily with *Minimal Search* while the latter obeys
the condition.

The current work seeks to capture a similar historical development, but
the analysis is replaced by a structure in which the target is assimilated
to the source: Both Latin and MR feature DET-categories in “SPEC” of
n. Crucially, in the current approach the DP-hypothesis is abandoned and

articles are phrasal. If the current analysis can be sustained, the claim that *Minimal Search* is the driving force behind diachronic change cannot be quite right, because DET-categories are invariably phrasal and there is no “grinding off” of specifiers to yield heads. While the forms and surface manifestations of DET-categories are subject to change, the key underlying change concerns n with ramifications in the DET-system: In the n_{wk} -languages of MR DET became obligatory to ensure a label.

Why did articles evolve and why did Latin lack them? The NSP predicts an obligatory XP in the sister position of $[_\alpha n_{wk} R]$ and the LBC. The NSP is consistent with the generalization given in (56) (cf. Uriagereka 1988; Corver 1989; and Bošković 2005a) – however, it does not strictly predict it:

- (56) Only languages without articles may allow LBE. (slightly modified from Bošković 2005a)

A possible route at an explanation of (56) is this: Articles diachronically emerge only in languages with n_{wk} , i.e., in complex nominals with an EPP-effect, perhaps analogous to expletives in the clause. For lexically designated expletives, there is evidence that they only indirectly relate to EPP-properties of different kinds and can evolve with a delay (e.g., cf. Fuß 2018 on the historical development the prefield expletive *es* ‘it’ to satisfy the verb second constraint in Middle High German). If similar diachronic paths can be established within complex nominals, the correlation between the availability of articles and the LBC could be captured: the nominal EPP is a precondition for the development of articles.

While I can only give this speculation here, it strikes me as a plausible and promising one. Notice that I interpret the obligatory presence of DET in n_{wk} -languages as an EPP-effect in the nominal domain. Articles can thus be considered a special strategy to fulfill the DET-requirement which n_{wk} gives rise to. As there are analogues in the clausal domain, such developments could be investigated in tandem.

5 Conclusion

The analysis I propose in this paper sheds new light on “Latin’s recourse to synthetic strategies, in contrast to those of a predominantly analytic nature in Romance” (Ledgeway 2012, 424): The historical development from a language with rich noun inflection (rich case paradigms and declension classes), i.e., Latin, to a set of languages with poor or no noun inflection, i.e., Modern Romance, is reflected in the Nominal Strength Parameter. While Merge applies freely, its output representations have to abide by the interface condition that every syntactic object be labeled by *Minimal Search*. This is met in the n_{str} -languages Latin without ado with concomitant optionality of determiners and the option of Left Branch Extraction. The Labeling Requirement can be met in n_{wk} -languages (modern Romance) only if $\{n, R\}$ is accompanied by xAP to yield $\{xAP, \{n, R\}\}$, deriving the obligatoriness of determiners and the effect of the Left Branch Condition. Loss of case morphology was thus a reflection of the parametric shift from n_{str} to n_{wk} , the underlying cause of rather dramatic syntactic changes in the nominal domain.

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